CHM 120

Observations of Chemical Changes

Final Report

Student Name	Ryan Saelens
Student ID	235544
Lesson	Observations of Chemical Changes
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Instructor	Jason Brinkley

Test Your Knowledge

Categorize each change as physical or chemical.

Physical Change	Chemical Change
1	2
Chopping tomatoes	Cooking tomatoes to make tomato sauce
Shredding paper	Burning paper
Peeling apples	Baking muffin batter to make muffins
Apple slices turning brown in the air	
Combining dry ingredients (flour, salt, sugar, baking powder) when making muffins	

Identify the products and the reactants in the reactions.

Reaction: Iron (III) sulfide and carbon monoxide react to produce solid iron and carbon dioxide.

Reactant	Product
1 Iron (III) sulfide	2 Iron (s)
Carbon monoxide	Carbon dioxide

Identify the products and the reactants in the reactions.

Reaction: Water and carbon dioxide are formed when ethane (C_2H_6) is burned in the presence of oxygen.

Reactant	Product
1 Ethane	2 Water
Oxygen	Carbon dioxide

Identify whether each statement is an observation or a conclusion.

Observation	Conclusion
1	2 Combustion occurred.
The reactants and products are clear.	No chemical reaction occurred.
The solution is cloudy and there are small flakes.	
The color changed from blue to green.	

Exploration

A chemical change may also be referred to as a _____.

- physical change
- chemical reaction
- chemical compound
- state of matter

Is silver tarnishing a chemical change or a physical change?

- Chemical change
- Physical change

A chemical equation is the written representation of a chemical reaction, showing the relationship between the molecules of the reactants (the starting chemicals) and the products (the new chemicals produced through the reaction).

- True
- False

An observation is _____.

- a description of what happened
- an explanation of why something happened
- a conclusion

Exercise 1

Suppose a household product label says it contains sodium hydrogen carbonate (sodium bicarbonate). Using your results from Data Table 1 as a guide, how would you test this material for the presence of sodium bicarbonate?

B I U 21 Word(s)

One possible way would be to drop HCl into a sample of sodium hydrogen carbonate and see if it fizzes up.

Write the chemical equation for the reaction in well A6.

B I U 0 Word(s)

One of the reactions you observed resulted in these products: NaCl + H_2O + CO_2 (g)? What well did this reaction occur in? Describe how the observations for this reaction support your answer.

B I U 27 Word(s)

A1

Being that the observations were fizzy and bubbly, it reminds me of a bottle or can of soda opening, releasing CO_2 and becoming bubbly and fizzy.

You found a sample of a solution that has a faint odor resembling vinegar (an acid). To verify that it is vinegar, you add a few drops of phenolphthalein. The sample turns pink. From this result, can you assume this sample is indeed vinegar or contains some vinegar? Explain your answer using your results from Data Table 1.

B I U 13 Word(s)

It would be hard to tell because phenolphthalein seems to turn things pink.

While performing a starch test on several different cookie brands, four tests result in the typical black color of starch presence, but the fifth gives a yellow-brown color. How might you interpret this result?

B I U 13 Word(s)

ItThe fifth test could have dealt with either potassium iodide or lead(II) nitrate

You have read that a new brand of hair tonic is supposed to contain lead (an ingredient in Grecian Formula®). Devise a simple test to confirm the presence or absence of lead in that hair tonic. Use your observations in Data Table 1 to describe a positive result for this test.

B I U 15 Word(s)

Add potassium iodide and if it turns yellow it is likely that it contains lead.

Is cutting a cake into 8 pieces a chemical or a physical change? Explain your answer.

B I U 3 Word(s)

A physical change

When a soda is poured into a glass and the soda bubbles, is it the result of a chemical change? Explain your answer.

B I U 9 Word(s)

No. The releasing of CO_2 is a physical change

In well B1, what was the reactant in the second step? Did this reactant cause a chemical change? Use your observations to support your answer.

B I U 2 Word(s)

The sunlight

Think about how phenolphthalein acts as an acid/base indicator. Do your observations in well A5 support this or contrast what you would expect to happen? Explain your answer.

B I U 11 Word(s)

It contrasts because I thought it would turn the solution pink.

Data Table 1: Chemical Change Observations

Well	Chemical #1 (4 drops)
A1	$NaHCO_3$ Sodium Bicarbonate
A2	IKI indicator
A3	KI Potassium Iodide
A4	NaOH Sodium Hydroxide
A5	HCl Hydrochloric Acid
A6	NaOH Sodium Hydroxide
B1	$AgNO_3$ Silver Nitrate
B2	NH_4OH Aqueous Ammonia

Well	Chemical #1 Appearance
A1	clear; thin liquid
A2	rust brown color; thick
A3	clear; somewhat sparkly
A4	clear
A5	clear
A6	clear
B1	clear; somewhat thick
B2	clear

Well	Chemical #2 (4 drops)
A1	HCl Hydrochloric Acid
A2	Starch
A3	$Pb(NO_3)_2$ Lead(II) Nitrate
A4	$C_{20}H_{14}O_4$ Phenolphthalein
A5	$C_{20}H_{14}O_4$ Phenolphthalein
A6	$AgNO_3$ Silver Nitrate
B1	NH_4OH Aqueous Ammonia
B2	$CuSO_4$ Copper(II) Sulfate

Well	Chemical #2 Appearance

A1	also clear
A2	clear; foggy
A3	clear
A4	clear
A5	clear
A6	clear
B1	clear; thin
B2	thin and darkish blue

Well	Observations
A1	became bubbly; somewhat fizzy
A2	black
A3	goldenrod color
A4	pink purple color
A5	very fizzy and foggy
A6	clumpy dark pieces in what looks like dirty, brown water
B1	Observation 1: sort of gray; foggy; + Absorb in paper towel and expose to sunlight Observation 2: looks like the paper got burnt somewhat; it's dirty-looking
B2	clear and sky blue mix; very marble looking

Well	Chemical Change (Yes/No)
A1	No.
A2	Yes.

A3	Yes.
A4	Yes.
A5	Yes.
A6	Yes.
B1	Yes.
B2	Yes.

Exercise 2

Describe the similarities and/or differences between heating and burning the magnesium metal. Did either heating or burning produce a chemical change? Explain your answer using the observations collected in Data Table 2.

B I U 28 Word(s)

The heating did nothing really. It just made it a little darker. The burning however had a chemical change. It burnt and gave off a very bright light.

Describe the similarities and/or differences between heating and burning the mossy zinc metal. Did either heating or burning produce a chemical change? Explain your answer using the observations collected in Data Table 2.

B I U 21 Word(s)

The heating just turned it an ash-like color. The burning melted it really fast. Therefore the burning involved a chemical change.

Describe the similarities and/or differences between heating and burning the $Cu(NO_3)_2$. Did either heating or burning produce a chemical change? Explain your answer using the observations collected in Data Table 2.

B I U 25 Word(s)

The heating turned it a green color. The burning resulted in the flame turning green. So the burning and heating both involved a chemical change.

Describe the similarities and/or differences between heating and burning the $CuCO_3$. Did either heating or burning produce a chemical change? Explain your answer using the observations collected in Data Table 2.

B I U 18 Word(s)

The heating essentially just burnt it up. Interestingly enough, the burning also turned the flame a green color.

How would you describe the differences between heating and burning? Use your experiences in the experiment to describe these differences.

B I U 64 Word(s)

The heating isn't direct contact, and due to the glass from the test tube, it can really only get so hot. When it is applied to the direct flame, there is more heat to take into consideration. This was seen with the magnesium lighting up upon contact with the direct flame, and the mossy zinc as it melted upon contact with the direct flame.

Which of the four chemicals that was heated produced a physical change? Support your answer with observations made and recorded in Data Table 2.

B I U 25 Word(s)

I'm thinking Mg because it only turned a little darker once heated, to where the other samples changed colors or burnt to a black crisp.

Data Table 2: Heating and Combustion

Chemical	Initial Observations	Heating Observations	Burning Observations
Mg	silver; rippled; very textured	a little darker shade of gray; nothing much happened	light up extremely bright
Zn	silver; sea shell roundish; looks like hard, thick tin foil	looks like the color of ash; kind of gray, kind of white	melted rather fast
$CuCO_3$	blue-green tiny particles; almost sand-like	it burnt quickly; black	turns the flame light green
$Cu(NO_3)_2$	dark blue particles; sand-like; akso looks like a blue, shattered crystal	started to bubble and change to a darker green color	turned the flame light green

Competency Review

A _____ is a change in the form of a substance.

- chemical change
- physical change

Identify an example of a chemical change.

- Boiling water
- Cooking an egg
- Condensing vapor
- Cutting a steak

A chemical reaction is a process where the atoms of substances interact with one another to form new substances.

- True
- False

In a chemical reaction the reactants are on the _____ of the written chemical equation.

- same side as the products
- right side
- left side

Physical observations can be used to determine if a chemical change has occurred.

- True
- False

A(n) _____ is the explanation of why something happened, and a(n) _____ is a description of what happened.

- observation; conclusion
- conclusion; observation

A white precipitate formed from the mixing of two liquids is an indication of a chemical change.

- True
- False

A metal is added to a liquid and bubbles form. What might be concluded?

- A chemical reaction occurred.
- A physical change occurred.
- Both a chemical and physical change occurred.

Heating chemicals is defined by the process of combustion.

- True
- False

_____ magnesium generates a very bright light.

- (selected) Burning
- Heating
- Both burning and heating

Extension Questions

Mark decided to investigate some household materials using the phenolphthalein solution. He placed 10 drops of each liquid or a small amount of solid (just enough to cover the tip of the spatula) in 2 mL of distilled water. Then he added 2 drops of the phenolphthalein solution and identified each substance as either an acid or a base, as shown in the table:

Tube #	Substance	Color before Phenolphthalein	Color after Phenolphthalein	Conclusion
1	Baking soda	Colorless	Pink	Base
2	White vinegar	Colorless	Colorless	Acid
3	Salt	Colorless	Colorless	Acid
4	Apple juice	Pale yellow	Pale yellow	Acid
5	Shampoo	Pale yellow	Pale yellow	Base
6	Window cleaner	Pale blue	Pale purple	Base
7	Milk	Slightly cloudy white	Slightly cloudy white	Acid

a. Can you draw any general conclusions about household materials based on his results?

b. Can you find any problems with Mark's logic? Explain.

(a) Based on his results, it seems like household materials are colored and are acids.

(b) Something seems off regarding his data for the apple juice and the shampoo. They have basically the same premises but different conclusions. I guess it just seems odd nonetheless.